$$I_0 = \frac{4x_0 - 2x_1}{R_0} = \frac{2}{R_0}\left(2\phi_0 - \phi_0\frac{q_{N-1}+q_1}{q_N}\right) = \frac{2\phi_0}{R_0}\left(2 - \frac{q_{N-1}+1}{q_N}\right) =$$
$$= \frac{2\phi_0}{R_0}\frac{2q_N - q_{N-1} - 1}{q_N} = \frac{2\phi_0}{R_0}\frac{2q_N - (2q_N - p_N) - 1}{q_N} = \frac{2\phi_0}{R_0}\frac{p_N - 1}{q_N}. \quad (26)$$

(5) $q_{N-1} = 2q_N - p_N$.

2.7 **[0.2점]**

$$U_0 = 2\varphi_0, \quad (27)$$

$$R_N = \frac{U_0}{I_0} = R_0\frac{q_N}{p_N - 1}. \quad (28)$$

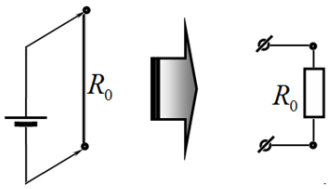
2.8 **[1.0점]** 1 .

2.

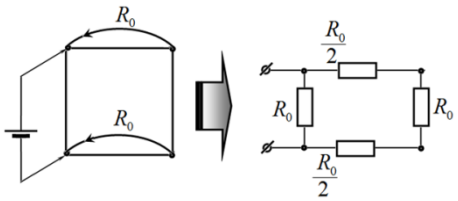
N	p_N	q_N	R_N
1	2	1	R_0
2	7	4	$R_0\frac{4}{7-1} = \frac{2}{3}R_0$
3	26	15	$R_0\frac{15}{26-1} = \frac{3}{4}R_0$
4	97	56	$R_0\frac{56}{97-1} = \frac{7}{12}R_0$
5	362	209	$R_0\frac{209}{362-1} = \frac{11}{19}R_0$

$N = 4$ 2.1 .

2.9 **[0.5점]** $N = 1$



$N = 2$



2 .

2.10 **[1.0점]** (28) .

$$p_N = \frac{1}{2}\left(\lambda^N - \lambda^{-N}\right), \quad q_N = \frac{1}{2\sqrt{3}}\left(\lambda^N + \lambda^{-N}\right), \quad (29)$$

$\lambda = 2 + \sqrt{3} > 1$.